QMS 050 Business Mathematics and Statistics

Module 1: Review of Basic Mathematics

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2026-11-04

Learning Outcomes

By the end of Module 1, you should be able to:

1. Perform operations on fractions confidently
2. Convert between fractions, decimals, and percentages, and apply percentages in business situations
3. Simplify and solve simple algebraic expressions and equations
4. Apply laws of powers and indices
5. Identify and solve problems involving arithmetic and geometric progressions

1.1 Fractions

What is a Fraction?

- A fraction represents a **part of a whole**
- Written as $\frac{a}{b}$, where:
  - $a =$ **numerator** (the part)
  - $b =$ **denominator** (the whole, cannot be 0)
- Example: If a business divides its budget into 4 equal parts and spends 1 part on rent, that is $\frac{1}{4}$ of the budget

Types of Fractions

- **Proper fraction**: numerator smaller than denominator, e.g. $\frac{3}{5}$
- **Improper fraction**: numerator larger than or equal to denominator, e.g. $\frac{9}{4}$
- **Mixed number**: whole number + fraction, e.g. $2\frac{1}{4}$

Converting improper → mixed:

$$\frac{9}{4} = 2\frac{1}{4} \quad \text{(since } 4 \times 2 = 8, \text{ remainder } 1\text{)}$$

Addition and Subtraction of Fractions

Rule: denominators must be the **same** (common denominator) before adding/subtracting.

Example 1:

$$\frac{1}{4} + \frac{1}{4} = \frac{1+1}{4} = \frac{2}{4} = \frac{1}{2}$$

Example 2 (different denominators):

$$\frac{1}{3} + \frac{1}{6}$$

Find common denominator (6): $\frac{1}{3} = \frac{2}{6}$

$$\frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

Worked Example: Subtraction

Example 3:

$$\frac{5}{6} - \frac{1}{3}$$

Step 1: Common denominator $= 6$

$$\frac{1}{3} = \frac{2}{6}$$

Step 2: Subtract

$$\frac{5}{6} - \frac{2}{6} = \frac{3}{6} = \frac{1}{2}$$

Business context: A shop sold $\frac{5}{6}$ of its stock, and $\frac{1}{3}$ of the stock was damaged before selling. What fraction was sold and undamaged? (Answer: $\frac{1}{2}$)

Multiplication of Fractions

Rule: multiply numerators together, multiply denominators together. No common denominator needed.

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

Example 4:

$$\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$$

Business context: A company allocates $\frac{2}{3}$ of its budget to operations, and $\frac{4}{5}$ of that operations budget goes to salaries. What fraction of the total budget is salaries? $\rightarrow \frac{8}{15}$

Division of Fractions

Rule: "Keep, Change, Flip" — keep the first fraction, change $\div$ to $\times$, flip (invert) the second fraction.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

Example 5:

$$\frac{3}{4} \div \frac{1}{2} = \frac{3}{4} \times \frac{2}{1} = \frac{6}{4} = \frac{3}{2} = 1\frac{1}{2}$$

Practice Questions 1.1 — Fractions

Try these on your own:

- (Q1) $\frac{1}{2} + \frac{1}{3}$
- (Q2) $\frac{3}{4} - \frac{1}{8}$
- (Q3) $\frac{2}{5} \times \frac{3}{7}$
- (Q4) $\frac{5}{6} \div \frac{2}{3}$
- (Q5) A warehouse has goods; $\frac{2}{5}$ are electronics and $\frac{1}{4}$ are furniture. What fraction is electronics and furniture combined?

1.2 Percentages

What is a Percentage?

- "Percent" means **"per hundred"**
- A percentage is a fraction with denominator 100
- Symbol: **%**

$$25\% = \frac{25}{100} = 0.25$$

Converting between forms:

Fraction	Decimal	Percentage
$\frac{1}{4}$	0.25	25%
$\frac{1}{2}$	0.5	50%
$\frac{3}{4}$	0.75	75%

Converting To and From Percentages

Fraction/decimal → percentage: multiply by 100

$$0.35 \times 100 = 35\%$$

Percentage → decimal: divide by 100

$$40\% = \frac{40}{100} = 0.4$$

Percentage → fraction: write over 100, then simplify

$$60\% = \frac{60}{100} = \frac{3}{5}$$

Finding a Percentage of a Quantity

Formula:

$$\text{Percentage of a number} = \frac{\text{Percentage}}{100} \times \text{Number}$$

Example 1: Find 20% of 5,000 TZS

$$\frac{20}{100} \times 5000 = 1000 \text{ TZS}$$

Example 2 (business): A shop gives a 15% discount on an item costing 40,000 TZS. What is the discount amount?

$$\frac{15}{100} \times 40000 = 6000 \text{ TZS}$$

Price after discount: $40000 - 6000 = 34000$ TZS

Percentage Increase and Decrease

Percentage Increase:

$$\text{New Value} = \text{Original} + (\% \times \text{Original})$$

Percentage Decrease:

$$\text{New Value} = \text{Original} - (\% \times \text{Original})$$

Example 3: A worker's salary of 500,000 TZS increases by 10%. Find the new salary.

$$500000 + \left(\frac{10}{100} \times 500000\right) = 500000 + 50000 = 550000 \text{ TZS}$$

Worked Example: Finding the % Change

Formula:

$$\%\text{ change} = \frac{\text{New} - \text{Old}}{\text{Old}} \times 100\%$$

Example 4: Sales rose from 200 units to 250 units. Find the percentage increase.

$$\%\text{ change} = \frac{250 - 200}{200} \times 100\% = \frac{50}{200} \times 100\% = 25\%$$

Example 5: Sales fell from 400 units to 350 units.

$$\%\text{ change} = \frac{350 - 400}{400} \times 100\% = \frac{-50}{400} \times 100\% = -12.5\%$$

(the negative sign shows a **decrease**)

Practice Questions 1.2 — Percentages

1. Convert $\frac{7}{20}$ to a percentage.
2. Find 35% of 12,000 TZS.
3. A shirt costs 25,000 TZS. It is discounted by 20%. Find the new price.
4. A company's revenue grew from 1,000,000 TZS to 1,250,000 TZS. Find the percentage increase.
5. An employee's allowance decreased from 80,000 TZS to 68,000 TZS. Find the percentage decrease.

1.3 Algebra

What is Algebra?

- Algebra uses **letters (variables)** like x, y, a to represent unknown numbers
- Useful in business to represent unknown quantities: cost, profit, number of units, etc.
- Example: If $x =$ number of items sold, and each item costs 2,000 TZS, then total revenue $= 2000x$

Basic Algebraic Terms

- **Term:** a single number, variable, or product, e.g. $5x$, 3, $2xy$
- **Expression:** combination of terms, e.g. $3x + 5$
- **Equation:** two expressions set equal, e.g. $3x + 5 = 20$
- **Coefficient:** the number multiplying a variable, e.g. in $7x$, coefficient is 7
- **Constant:** a term with no variable, e.g. in $3x + 5$, 5 is the constant

Simplifying Algebraic Expressions

Rule: only "like terms" (same variable and power) can be combined.

Example 1:

$$3x + 5x = 8x$$

Example 2:

$$7x + 3y - 2x + y = (7x - 2x) + (3y + y) = 5x + 4y$$

Example 3 (with brackets):

$$2(x + 3) = 2x + 6$$

$$3(2x - 1) + 4 = 6x - 3 + 4 = 6x + 1$$

Your Turn: Try these

Rules to remember: Combine *like terms* (same variable and power) and distribute fully across brackets.

- **Q1** Simplify $4x + 9x - 2x$
- **Q2** Simplify $6x + 8y - 3x - 5y$
- **Q3** Simplify $5a + 7 - 2a + 4$
- **Q4** Simplify $4(2x - 3)$
- **Q5** Simplify $2(3x + 4) + 5x - 2$

1.4 Powers and Indices

What are Powers/Indices?

- Power (index/exponent) shows **how many times a number is multiplied by itself**

$$a^n = a \times a \times a \times \cdots \times a \quad (n \text{ times})$$

- $a =$ base, $n =$ power/index/exponent

Example:

$$2^4 = 2 \times 2 \times 2 \times 2 = 16$$

Laws of Indices (Part 1)

Law 1 — Multiplication: add the powers

$$a^m \times a^n = a^{m+n}$$

Example: $2^3 \times 2^2 = 2^{3+2} = 2^5 = 32$

Law 2 — Division: subtract the powers

$$a^m \div a^n = a^{m-n}$$

Example: $5^6 \div 5^2 = 5^{6-2} = 5^4 = 625$

Law 3 — Power of a power: multiply the powers

$$(a^m)^n = a^{m \times n}$$

Example: $(3^2)^3 = 3^6 = 729$

Laws of Indices (Part 2)

Law 4 — Zero power: any number (except 0) to power $0 = 1$

$$a^0 = 1 \quad \text{e.g. } 9^0 = 1$$

Law 5 — Negative power: means reciprocal

$$a^{-n} = \frac{1}{a^n} \quad \text{e.g. } 2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

Law 6 — Fractional power: means root

$$a^{\frac{1}{n}} = \sqrt[n]{a} \quad \text{e.g. } 8^{\frac{1}{3}} = \sqrt[3]{8} = 2$$

Worked Examples: Indices

Example 1: Simplify $3^4 \times 3^{-2}$

$$3^{4+(-2)} = 3^2 = 9$$

Example 2: Simplify $\dfrac{x^7}{x^3}$

$$x^{7-3} = x^4$$

Example 3: Evaluate $16^{\frac{1}{2}}$

$$16^{\frac{1}{2}} = \sqrt{16} = 4$$

Business context: Powers are used in **compound interest**: if an investment grows at rate r for n years, the amount is $A = P(1+r)^n$

Practice Questions 1.4 — Powers and Indices

1. Simplify: $4^2 \times 4^3$
2. Simplify: $6^5 \div 6^2$
3. Simplify: $(2^3)^2$
4. Evaluate: 5^0
5. Evaluate: 2^{-4}
6. Evaluate: $27^{\frac{1}{3}}$

1.5 Sequences and Series

What is a Sequence?

- A **sequence** is an ordered list of numbers following a rule
- Each number in the sequence is called a **term**
- A **series** is the **sum** of the terms of a sequence
- Example sequence: 2, 4, 6, 8, 10 ...
- Example series: $2 + 4 + 6 + 8 + 10$

Arithmetic Progression (AP)

- A sequence where each term is obtained by **adding (or subtracting) a constant** value, called the **common difference (d)**, to the previous term

Example: 3, 7, 11, 15, 19 ... (here $d = 4$)

nth term formula:

$$A_n = a_1 + (n-1)d$$

where a_1 = first term, d = common difference, n = term position

Worked Example: AP — Finding a Term

Example 1: Find the 10th term of the AP: 5, 8, 11, 14, ...

Here $a_1 = 5$, $d = 8 - 5 = 3$, $n = 10$

$$A_{10} = a_1 + (n-1)d = 5 + (10-1)(3) = 5 + 27 = 32$$

Business context: A company's sales increase by a fixed 50 units every month, starting at 200 units in month 1. Find sales in month 6.

$$a_1 = 200,\ d = 50,\ n = 6$$

$$A_6 = 200 + (6-1)(50) = 200 + 250 = 450 \text{ units}$$

Sum of an Arithmetic Progression

Formula:

$$S_n = \frac{n}{2}[2a_1 + (n-1)d]$$

Example 2: Find the sum of the first 10 terms of AP: 5, 8, 11, 14, ...

$$a_1 = 5,\ d = 3,\ n = 10$$

$$S_{10} = \frac{10}{2}[2(5) + (10-1)(3)] = 5[10 + 27] = 5(37) = 185$$

Geometric Progression (GP)

- A sequence where each term is obtained by **multiplying** the previous term by a constant value, called the **common ratio (r)**

Example: 2, 6, 18, 54, ... (here $r = 3$)

nth term formula:

$$G_n = a_1 \, r^{n-1}$$

where a_1 = first term, r = common ratio, n = term position

Worked Example: GP — Finding a Term

Example 3: Find the 6th term of the GP: 3, 6, 12, 24, ...

$a_1 = 3$, $r = 6 \div 3 = 2$, $n = 6$

$$G_6 = a_1\, r^{n-1} = 3(2)^5 = 3 \times 32 = 96$$

Business context: An investment of 100,000 TZS grows at 10% per year (i.e. multiplied by 1.1 each year). Find the value after 4 years — this is G_5 (year 0 to year 4, i.e., 5 terms including the start).

$$a_1 = 100000,\ r = 1.1$$

$$G_5 = 100000(1.1)^4 \approx 146410 \text{ TZS}$$

Sum of a Geometric Progression

For $r = 1$: $\quad S_n = a_1 r$

Formula (for $r \neq 1$):

$$S_n = \frac{a_1(r^n - 1)}{r - 1} \quad \text{(useful when } r > 1\text{)}$$

or equivalently

$$S_n = \frac{a_1(1 - r^n)}{1 - r} \quad \text{(useful when } r < 1\text{)}$$

Example 4: Find the sum of the first 5 terms of GP: 3, 6, 12, 24, ...

$$a_1 = 3,\ r = 2,\ n = 5$$

$$S_5 = \frac{3(2^5 - 1)}{2 - 1} = \frac{3(31)}{1} = 93$$

Practice Questions 1.5 — Sequences & Series

1. Find the 8th term of AP: 4, 9, 14, 19, ...
2. Find the sum of the first 12 terms of AP: 2, 5, 8, 11, ...
3. Find the 5th term of GP: 4, 8, 16, 32, ...
4. Find the sum of the first 6 terms of GP: 2, 6, 18, 54, ...
5. A worker's salary starts at 300,000 TZS and increases by 20,000 TZS every year (AP). Find the salary in year 5.

References

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- Tuttle, M., Graves, V., Rouche, N. (2005). *Business Mathematics*, 9th Edition. Prentice Hall.
- Soper, J. (2004). *Mathematics for Economics and Business: An Interactive Introduction*, 2nd Edition. Blackwell Publishing Professionals.

Review Questions — Module 1

Section A: Short Answer

- Q1. Simplify $\frac{2}{3} + \frac{1}{6} - \frac{1}{4}$
- Q2. A phone costing 300,000 TZS is discounted by 12%. Find the sale price.
- Q3. Solve for x: $5x - 4 = 21$
- Q4. Simplify $\frac{a^5 \times a^2}{a^3}$
- Q5. Find the 15th term of the AP: 7, 11, 15, 19, ...

Section B: Applied Problems

- Q6. A company's revenue was 8,500,000 TZS last year and rose to 9,775,000 TZS this year. Find the percentage increase.
- Q7. Total cost of production is given by $C = 800x + 15000$. Find the number of units x produced if the total cost is 55,000 TZS.
- Q8. A savings account starts with 50,000 TZS and earns interest such that the balance is multiplied by 1.08 every year. Find the balance after 3 years (i.e., the 4th term of the GP, including year 0).
- Q9. Find the sum of the first 20 terms of the AP: 3, 6, 9, 12, ...
- Q10. Evaluate: $9^{\frac{1}{2}} \times 9^2$

End of Module 1